Multi-Agent Reinforcement Learning with MAPPO

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1 Introduction

Multi-agent reinforcement learning introduces unique challenges in cooperative environments, particularly when agents must coordinate complex, interdependent actions in environments with sparse rewards. The Overcooked-AI benchmark exemplifies such challenges by simulating constrained kitchen environments where two agents must collaborate to efficiently prepare and deliver soups. This project investigates the use of a centralized training and decentralized execution (CTDE) framework to enhance coordination between agents. The Multi-Agent Proximal Policy Optimization (MAPPO) algorithm was selected for its stability and effectiveness in multi-agent continuous control tasks. The key design choices include environment-agnostic reward shaping and structured hyperparameter tuning to promote generalization across diverse kitchen layouts. Rather than relying solely on sparse end-goal rewards such as soup delivery, the learning process integrates shaped rewards for intermediary actions like ingredient collection and dish handling, which guide agents toward coordinated behavior. The training setup is kept consistent across all environments to ensure fair comparison and meaningful analysis of generalization and collaboration.

2 Problem Overview and Environment Setup

The task involves training two cooperative agents (chefs) to maximize the number of soups successfully delivered in a simulated Overcooked kitchen. Each episode spans a fixed 400-timestep horizon, during which agents must complete a structured sequence of actions—collecting onions, placing them into pots, waiting for the soup to cook, retrieving dishes, picking up the soup, and delivering it to the serving station. A complete soup delivery yields a +20 reward, while incomplete or invalid attempts offer no reward. The environment is fully deterministic, meaning that for a given policy, the sequence of observations, actions, and rewards remains identical across episodes.The agents are trained and evaluated across three predefined layouts of increasing difficulty—**cramped_room**, **coordination_ring**, and **counter_circuit_o_1order**—which vary in navigational constraints and the degree of cooperation required. All layouts share the same state representation and action space, in addition to the same hyperparameter for training enabling consistent policy design and reward shaping strategies across environments.

2.1 State and Action Space

The Overcooked environment is modeled as a fully observable Markov Decision Process (MDP), where each agent receives a 96-dimensional observation vector that captures a player-centric view of the environment. These observations are generated specifically for each agent and include spatial and contextual information. The feature vector encodes the agent's orientation, the object it is currently holding, distances to relevant game elements such as onions, dishes, pots, soups, and the delivery station, as well as the state of nearby pots including their cooking status and ingredient count. It also includes the agent's proximity to walls, the relative position of the other agent, and each agent's absolute position on the map. This comprehensive observation allows agents to make context-aware decisions that depend not only on their own state but also on the spatial dynamics of the other agent and the overall kitchen environment. The action space available to each agent is discrete and consists of six possible actions: move up, down, left, or right; stay in place; and interact. The 'interact' action is context-sensitive, enabling tasks like picking up items, placing ingredients in pots, or delivering soups based on the tile the agent is facing. This small set of actions, along with the state information, is enough for the agents to learn how to work together effectively in all layouts.

2.2 Reward Function

The default Overcooked environment provides a sparse reward of +20 for each successful soup delivery. However, this reward is difficult to attain during early training since it requires agents to execute a sequence of interdependent actions spanning at least 20 timesteps. To address the challenge posed by this sparse signal, reward shaping is introduced to reinforce incremental progress toward task completion. The shaped reward function assigns intermediate rewards to key subtasks in the soup preparation pipeline: placing onions into pots (+3), picking up dishes (+3), and collecting cooked soup (+5). Additionally, a small proximity-based reward (+0.1) encourages agents to reduce the distance between a held dish and the delivery station. During training, the sparse soup delivery reward (+20) is evenly divided between the two agents (+10 each), aligning with the centralized training setup where cooperative success is jointly reinforced. The final per-agent shaped reward is computed as the sum of the shaped subtask reward and half of the sparse soup reward when applicable. This reward structure promotes the discovery of useful cooperative behaviors such as ingredient gathering, pot preparation, and dish management, even before full delivery is achieved. By rewarding intermediate progress, the agents are better guided through the task structure, leading to more efficient learning and enhanced coordination across all layouts.

3 Methodology: Centralized Training with MAPPO

The primary objective of this project is to enable robust collaboration between two agents across diverse Overcooked layouts by leveraging a unified training strategy that generalizes well to varying spatial constraints and task dependencies. The project focuses specifically on improving coordination between agents without relying on layout-specific tuning or asymmetric architectures. The hypothesis is that by using centralized training with decentralized execution, shared reward signals, and layout-agnostic hyperparameters, agents can learn generalizable cooperative behaviors that transfer effectively across different environments. This approach aims to address key challenges in multi-agent reinforcement learning—such as credit assignment and inter-agent coordination—within a unified framework. To solve the Overcooked multi-agent cooperation challenge across all three layouts—cramped_room, coordination_ring, and counter_circuit_o_1order—a Multi-Agent Proximal Policy Optimization (MAPPO) algorithm was implemented. The motivation behind selecting MAPPO lies in its stability and sample efficiency in cooperative multi-agent settings, as well as its capacity to handle shared rewards and decentralized execution effectively. MAPPO enables agents to be trained with shared critic feedback during centralized training, while still acting independently during deployment. The policy architecture is based on an actor-critic model. Each agent's actor network generates a probability distribution over discrete actions, while the critic estimates a scalar value function for advantage calculation. The networks share a common architecture comprising two fully connected layers of 256 units each, with GELU activations and Layer Normalization to improve learning stability. The action space consists of six discrete actions—move up, down, left, right, stay in place, and a context-sensitive interact. Both agents share the same network architecture and hyperparameters, ensuring consistent policy updates and symmetry across all layouts. To reduce high variance in gradient estimates and improve sample efficiency, Generalized Advantage Estimation (GAE) is used with a discount factor $\gamma = 0.90$ and a smoothing factor $\lambda = 0.95$. The γ parameter controls how far into the future the agent considers rewards, while λ balances the bias-variance tradeoff when estimating advantages. For policy updates, the PPO surrogate objective is clipped using $\epsilon = 0.01$, preventing large policy shifts and ensuring stable learning. Entropy regularization is applied to encourage early-stage exploration and prevent premature convergence to suboptimal deterministic policies. Normalization plays a critical role in generalization across layouts. Given the 96-dimensional agent-centric observations, input normalization is applied online by maintaining a running mean and variance of observed features, allowing the agent to adapt to changing input distributions during training. This step ensures that agents are not sensitive to varying feature scales that arise due to differences in layout geometry or item placement. Reward shaping is another integral component of the methodology. To address the challenge of sparse rewards in Overcooked, a shaped reward structure was used to reinforce incremental task progress. These intermediate re-

wards support the learning of coordinated subtasks necessary for successful soup delivery. A full description of the shaped reward function is provided in the Reward Function design (section 2.2). This structure encourages the agents to learn sub-tasks essential for delivering soup, which in turn leads to emergent collaboration. During training, agents sample actions from their learned policy distributions to ensure sufficient exploration. During evaluation, two strategies were considered for action selection: greedy (deterministic) and stochastic sampling. In the greedy approach, agents always selected the action with the highest probability from the policy distribution. While this method was intended to provide a stable and consistent measure of policy performance, it frequently led to suboptimal results—particularly in the counter_circuit_o_1order layout, where narrow passages and spatial bottlenecks required agents to dynamically adapt their paths and actions. The absence of stochasticity often caused agents to fall into repetitive patterns or deadlocks. In contrast, sampling from the policy distribution during testing allowed for more varied behavior and resulted in improved coordination. As a result, final evaluations were conducted using stochastic policies over 100 consecutive test episodes per layout, with metrics such as average soups delivered, onion pickups, and dish pickups recorded. In summary, the methodology integrates centralized training, shared policy networks, shaped rewards, GAE, normalization, and PPO updates to foster collaboration between agents and achieve generalization across all three Overcooked layouts. Empirical results validate the hypothesis, demonstrating effective coordination without layout-specific modifications.

3.1 Training Process

The training pipeline is designed to develop a generalizable collaborative policy across all three Overcooked layouts. A shared MAPPO-based framework is used to train two agents cooperatively without layout-specific customization. Each layout is trained independently using the same policy architecture, reward shaping scheme, and hyperparameter configuration. Layout-specific training durations are set to 35,000 episodes for the first two layouts and 50,000 episodes for the more complex `counter_circuit_o_1order`. Each training episode runs for a fixed horizon of 400 steps. During rollout, agents interact with the environment using a stochastic policy, sampling actions from the actor's probability distribution. The training loop collects trajectories containing states, actions, log probabilities, critic value estimates, and rewards. These transitions are stored temporarily and processed at the end of each episode. The critic predicts state values at every timestep, which are used in conjunction with actual rewards to compute the advantage estimates. GAE is employed to calculate the temporal advantage of each action using the critic's predictions and observed rewards. This allows the agent to weigh short-term and long-term effects of its actions using discounted future returns, controlled by parameters $\gamma = 0.90$ and $\lambda = 0.95$. Once GAE and target returns are computed, policy updates are performed using Proximal Policy Optimization (PPO). The update loop processes the trajectory data through a PyTorch `DataLoader`, applying multiple PPO epochs over minibatches. The PPO loss includes a clipped surrogate objective to stabilize updates, a mean-squared value loss for the critic, and an entropy bonus to promote exploration. The total loss gradient is backpropagated through the shared actor-critic network, which is composed of two hidden layers and shared feature extraction layers. Gradient clipping is applied with a norm threshold to prevent instability during large updates. State normalization is integrated into the training loop via online tracking of running mean and variance statistics. All observations are normalized before being passed into the network, ensuring stability across layouts with different spatial scales or object distributions. The shaped reward structure plays a vital role in guiding learning. In addition to the sparse +20 reward for delivering soup, intermediate rewards are added for placing ingredients into pots (+3), picking up dishes (+3), collecting soup (+5), and a small shaping reward (+0.1) for reducing dish-to-delivery distance. These rewards are split evenly between agents to reflect shared task responsibility, encouraging synchronized behaviors that naturally emerge through optimization. Performance is evaluated by running 100 consecutive test episodes and metrics such as soups delivered, dish pickups, and onion pickups are logged and visualized through moving averages over time. These metrics provide insight into how well collaboration has emerged, as efficient teamwork leads to higher soup throughput and coordinated task completion. This pipeline ensures that all components—policy architecture, reward shaping, normalization, GAE, PPO, and evaluation—contribute

toward the overarching objective of enabling decentralized but collaborative behavior across multiple environments.

3.2 Choosing MAPPO

Multi-Agent Proximal Policy Optimization (MAPPO) was chosen due to its proven effectiveness in handling cooperative multi-agent environments, particularly those involving shared rewards and coordinated actions from decentralized observations. MAPPO extends Proximal Policy Optimization (PPO) to support centralized training and decentralized execution, directly addressing the coordination and credit-assignment challenges inherent in Overcooked. Alternative methods, such as Value Decomposition Networks (VDN), were initially explored but failed to yield meaningful results, potentially due to implementation complexity, insufficient modeling of agent interdependencies, or limitations inherent to the approach in tightly coupled tasks. By contrast, MAPPO demonstrated stability, efficiency, and suitability to Overcooked's cooperative requirements. However, MAPPO exhibited some limitations. Shaped intermediate rewards, although beneficial for guiding learning, occasionally led to unintended agent behaviors, such as excessive prioritization of sub-tasks like dish pickups. Furthermore, achieving consistent performance across all layouts was challenging due to varying complexity levels; notably, the counter_circuit_o_1order layout required substantially more training episodes to attain adequate performance, underscoring the limitations of a fixed hyperparameter configuration. Future work may investigate curriculum learning, adaptive reward shaping, or architectures incorporating temporal or historical context to further refine coordination and training efficiency.

4 Experiments and Results

To evaluate the performance and generalizability of the MAPPO-based multi-agent solution, experiments were conducted across the three cramped_room, coordination_ring and counter_circuit_o_1order Overcooked layouts. Each layout introduces unique coordination challenges, requiring different levels of agent collaboration and task sequencing. Two agents were trained using a centralized training strategy for each layout, with the algorithm, reward shaping configuration, and hyperparameter settings held constant across all environments. While the learning procedure and network architecture remained unchanged, the number of training episodes varied to accommodate layout-specific complexities. The more demanding counter_circuit_o_1order layout was allocated 50,000 training episodes, while the others were trained for 35,000 episodes each.

Agent performance during training was monitored using the following metrics:

- **Number of soups delivered per episode**: Directly measures the agents' cooperative effectiveness in achieving the primary task objective.
- **Frequency of onion pickups**: Indicates the agents' proficiency in performing essential preparatory tasks, reflecting effective resource collection and coordination.
- **Frequency of dish pickups**: Reflects how effectively agents manage necessary resources for final delivery, providing insight into subtask management and collaborative efficiency.
- **Rolling averages computed every 100 episodes**: Smooths episodic variance, enabling clearer interpretation of learning trends over time.

For evaluation, each trained agent was tested over 100 consecutive episodes. Although greedy action selection—choosing the most probable action—was initially considered, empirical results showed that sampling actions from the policy distribution yielded better coordination and higher task performance. As a result, the stochastic policy was retained during evaluation. The same metrics collected during training were recorded during testing to enable consistent and interpretable comparisons. To

ensure fair reporting and reproducibility, reward shaping parameters, training architecture, and hyperparameters were fixed across all experiments. A constant random seed (`SEED = 42`) was used for all training and evaluation runs.

4.1 Hyperparameter Selection

All hyperparameters were uniformly applied across the three Overcooked layouts to satisfy project constraints and support generalization. Initial values were selected based on commonly adopted MAPPO configurations and were subsequently refined based on empirical observations during early training stages. Performance improved substantially when the discount factor γ was reduced from 0.99 to 0.90. This adjustment decreased the emphasis on long-term rewards, making agents more sensitive to short-term cooperative benefits, which aligns better with the 400-timestep episodes in Overcooked. In addition, reducing the PPO clipping threshold ϵ from 0.2 to 0.01 helped stabilize training by constraining policy updates and avoiding large deviations. The learning rate was set to 3×10^{-4}, and Generalized Advantage Estimation (GAE) was applied with $\lambda = 0.95$ to manage the bias-variance trade-off. Entropy regularization was used with a coefficient of 0.01 to promote early-stage exploration. The value loss was weighted by a critic coefficient of 0.5, and gradient updates were clipped to a maximum norm of 0.5 to maintain numerical stability. Each PPO update consisted of three epochs with a batch size of 256. No layout-specific hyperparameter tuning was conducted, and all configurations were fixed throughout to maintain consistency and comparability across environments.

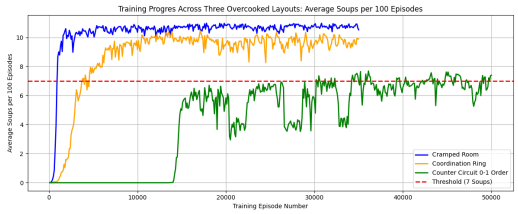
4.2 Results

The results presented in this section highlight the performance of the MAPPO-based multi-agent policy across the three Overcooked layouts: cramped_room, coordination_ring, and counter_circuit_o_1order. All environments were evaluated under the same hyperparameter settings and network architecture, isolating the effects of layout complexity and coordination requirements. Performance was evaluated based on the number of soups delivered during both training and testing phases. In addition to the primary objective, intermediate task-related metrics such as onion pickups and dish pickups were tracked to provide further insights into how well agents internalized sub-tasks and how reward shaping influenced learning dynamics.

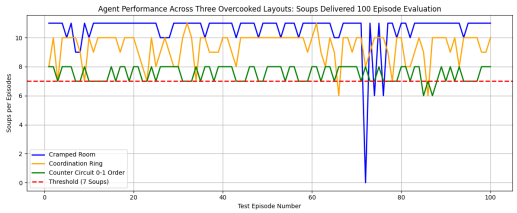
Training Performance Across Layouts

Training was conducted over >35000 episodes, during which agents progressively improved in their ability to deliver soups. The average number of soups delivered per 100 episodes increased steadily for all layouts. As seen in the training curve (Figure 1 (a)), cramped_room agents achieved the highest performance, consistently surpassing the success threshold of 7 soups per episode. Coordination_ring also demonstrated strong learning dynamics, though with more fluctuations and delayed stability. Counter_circuit_o_1order, the most complex layout, required longer training and demonstrated slower and more limited progress, ultimately stabilizing just above the success threshold with noticeably lower final performance compared to the other two layouts. Intermediate task metrics, such as dish pickups (Figure 2 (c)) and onion pickups (Figure 2 (a)), provide deeper insights into behavioral patterns across layouts. Agents in the cramped_room environment consistently showed higher dish pickup frequency throughout training, reflecting efficient coordination and smooth cycling through sub-tasks. In contrast, dish pickups in the counter_circuit_o_1order layout exhibited significant variability, not stabilizing until later training stages, indicating initial difficulties in task management due to the layout's complexity and spatial constraints. However, onion pickups for the counter_circuit_o_1order layout were notably higher and increased steadily during later training phases. This suggests that, while agents faced challenges with consistently managing dishes due to the complex pathways and bottlenecks in this layout, they successfully optimized onion collection strategies, potentially as a compensatory behavior. Throughout training, overall agent performance across all three layouts improved steadily, as evidenced by rising soup deliveries, peaking near or

above the success threshold of 7 soups per episode. Nonetheless, distinct subtask execution patterns highlight how agents adapt differently based on environmental complexity and task dynamics.

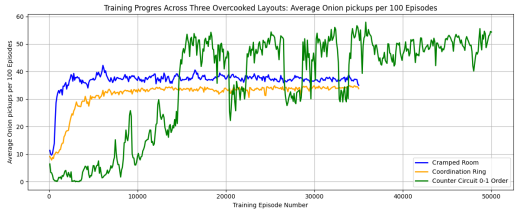


(a) Training Performance: Soups Delivered

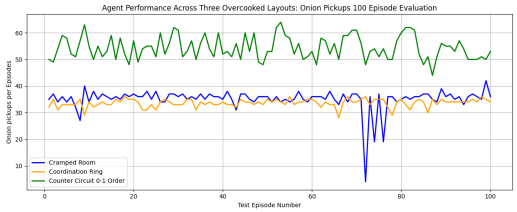


(b) Test Performance: Soups Delivered

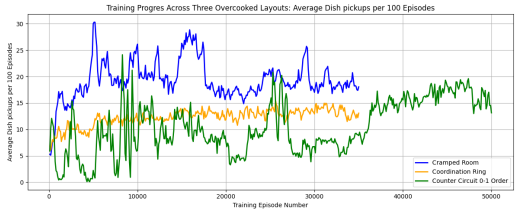
Figure 1: Avg Soups delivered during Training and Testing across all layouts



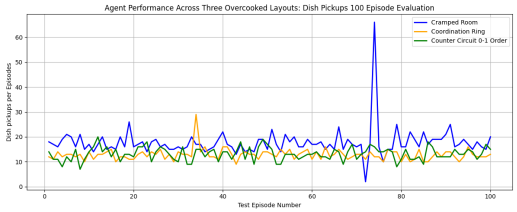
(a) Training Performance: Onion Pickups



(b) Test Performance: Onion Pickups



(c) Training Performance: Dish Pickups



(d) Test Performance: Dish Pickups

Figure 2: Avg Onion and Dish Pickups during Training and Testing across all layouts

Test Performance and Generalization To evaluate policy generalization and performance, each trained model was run for 100 consecutive test episodes, using stochastic action sampling from the policy distribution rather than greedy action selection. This approach introduced variability in action choices, reducing the risk of repetitive deadlock patterns that can occur in deterministic environments. The results (Table 1) showed that agents were generally capable of maintaining learned coordination strategies during deployment across all layouts. The cramped_room layout produced the highest performance, averaging approximately 10.6 soups per episode. This suggests that simpler spatial constraints and symmetrical layout features significantly support robust collaborative behaviors. The coordination_ring environment, though achieving strong overall performance, displayed slightly lower average deliveries (9.36 soups per episode) and greater variability. This can be attributed to increased coordination complexity and potential bottlenecks requiring dynamic agent adjustments. Performance in the counter_circuit_o_1order layout was just above the success threshold (7.51 soups per episode), clearly reflecting its higher coordination difficulty due to complex pathways and constrained maneuverability. Despite lower absolute numbers, performance steadily improved through extended training, suggesting the agents gradually adapted to its complexity. An interesting behavioral observation during testing was a noticeable dip in performance for the cramped_room layout around episode 70. This sharp decline coincided with increased dish pickup frequency and decreased onion pickups and soup deliveries, suggesting that agents momentarily prioritized suboptimal intermediate tasks. Such behavior illustrates potential pitfalls in reward shaping, where excessive emphasis on intermediate actions may occasionally divert agents from the primary

objective of soup delivery. Overall, the results demonstrate the effectiveness of the MAPPO algorithm with shaped rewards and stochastic action sampling in fostering reliable coordination across diverse environmental complexities. The unified approach succeeded in maintaining consistently strong performance across all tested layouts without layout-specific tuning.

Layout	**Avg. Soups Delivered**
cramped_room	10.60
coordination_ring	9.36
counter_circuit_o_1order	7.51

Table 1: Avg soups delivered over 100 consecutive test episodes for all layouts

5 Ablation Study

An ablation study was conducted to isolate and evaluate the contributions of critical components within the MAPPO-based learning pipeline. Due to computational limitations, the analysis was not exhaustive but focused on key architectural and training choices that were hypothesized to significantly impact learning stability and performance. The removal of shaped rewards—leaving only the sparse +20 reward for soup delivery—resulted in a complete failure to learn effective policies. Without intermediate feedback, agents were unable to discover the long sequence of actions required for a successful delivery, highlighting the necessity of shaped rewards such as those for placing onions, picking up dishes, or retrieving cooked soup. Adjustments to the PPO clipping threshold also yielded significant differences. An initial value of $\epsilon = 0.2$ led to unstable learning and erratic policy updates. Lowering this value to $\epsilon = 0.01$ produced more gradual and reliable convergence, particularly in layouts requiring precise coordination, suggesting that smaller clipping ranges are beneficial in highly structured multi-agent environments. Disabling observation normalization introduced instability during training. Raw input features—especially spatial distances and one-hot encoded states—varied greatly in scale, leading to inefficient learning. Re-enabling running mean and variance normalization corrected this issue, demonstrating that input scaling is essential for generalization across diverse layout geometries. Finally, evaluation was performed using both greedy (deterministic) and stochastic (sampling-based) action selection. While greedy evaluation provided consistent behaviors, it occasionally led to coordination deadlocks, particularly in tight layouts. Stochastic policies maintained a degree of flexibility, which improved performance in such scenarios. This outcome suggests that retaining some level of policy entropy, even during testing, may benefit environments with high inter-agent interference.

6 Pitfalls and Challenges

Implementing the algorithms from scratch presented significant initial challenges. The initial approach employed Value Decomposition Networks (VDN), but despite multiple attempts and extensive adjustments, the implementation failed to achieve meaningful performance—agents consistently struggled to deliver even a single soup. This lack of progress highlighted potential implementation inaccuracies or inherent limitations in the VDN approach within this specific problem context. Consequently, the shift was made to Multi-Agent Proximal Policy Optimization (MAPPO). The MAPPO implementation evolved incrementally, starting from basic actor-critic setups and gradually integrating advanced techniques such as Generalized Advantage Estimation and input normalization. Each addition required thorough research, multiple debugging iterations, and validation through experimental runs. Hyperparameter tuning posed another considerable challenge due to the sheer number of parameters involved. Parameters such as the learning rate, PPO clipping threshold, and entropy coefficient were highly sensitive, with minor adjustments often resulting in significant variations in performance. Achieving consistent and reliable results necessitated extensive trial and error, as well as careful documentation of performance outcomes across numerous experimental setups. Another major obstacle was the inherent sparsity of rewards within the Overcooked environment. The sparse

reward structure, which provides a +20 reward only upon successful soup delivery, required agents to execute prolonged sequences of coordinated actions. Without shaped intermediate rewards, agents frequently failed to receive sufficient learning signals, significantly impeding their ability to discover effective collaborative strategies. To address this, a reward shaping mechanism was designed to reinforce critical subtasks, guiding agents progressively towards successful task completion. Inter-agent coordination also presented considerable difficulties. In all of the layouts, successful task completion demanded precise synchronization and adaptive role assignment among agents. The agents had to implicitly learn coordination solely through environmental feedback, often resulting in inconsistent and suboptimal strategies, especially during early training phases. Another significant challenge was the substantial number of training episodes required for the agents to reach acceptable performance, particularly in the more complex counter_circuit_o_1order layout. While simpler layouts such as cramped_room and coordination_ring demonstrated noticeable improvement within approximately 15,000 episodes, the counter_circuit_o_1order environment required at least 50,000 episodes to consistently surpass the success threshold of 7 soups per episode. This highlights the increased complexity and coordination demands of certain layouts, and the corresponding necessity for prolonged training, further emphasizing the computational expense and inefficiency associated with training agents from scratch without curriculum learning or transfer learning methods. Finally, the deterministic nature of the environment complicated agent evaluation. Deterministic (greedy) policies, while ensuring reproducibility, occasionally resulted in fragile behaviors prone to performance drops caused by repeated conflicts or deadlocks. This issue was mitigated by evaluating agents using stochastic action selection, introducing variability and reducing the likelihood of persistent coordination failures.

7 Conclusion and Future Work

This project successfully explored multi-agent cooperative behavior using the Multi-Agent Proximal Policy Optimization (MAPPO) algorithm in the challenging Overcooked environment. The implemented solution effectively fostered coordination between agents through centralized training and decentralized execution, supported by strategic reward shaping and input normalization. Experiments demonstrated the agents' ability to generalize across diverse layouts, confirming the effectiveness of the adopted approach for cooperative task scenarios. However, several opportunities for improvement remain. Not all layouts achieved optimal performance, and episodes of coordination breakdown indicated the need for richer information processing. Future work should consider more advanced multi-agent reinforcement learning algorithms such as Value Decomposition Networks (VDN) or QMIX, which explicitly model joint action values and may better handle complex cooperation dynamics. Incorporating temporal context through techniques like state stacking or recurrent neural networks could also provide agents with historical information, enabling more informed and adaptive decision-making. Additionally, leveraging curriculum learning or transfer learning approaches could streamline training processes, enhance generalization, and mitigate hyperparameter sensitivity. These avenues promise significant advancements in multi-agent reinforcement learning, facilitating robust, scalable, and adaptive cooperative behavior.

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